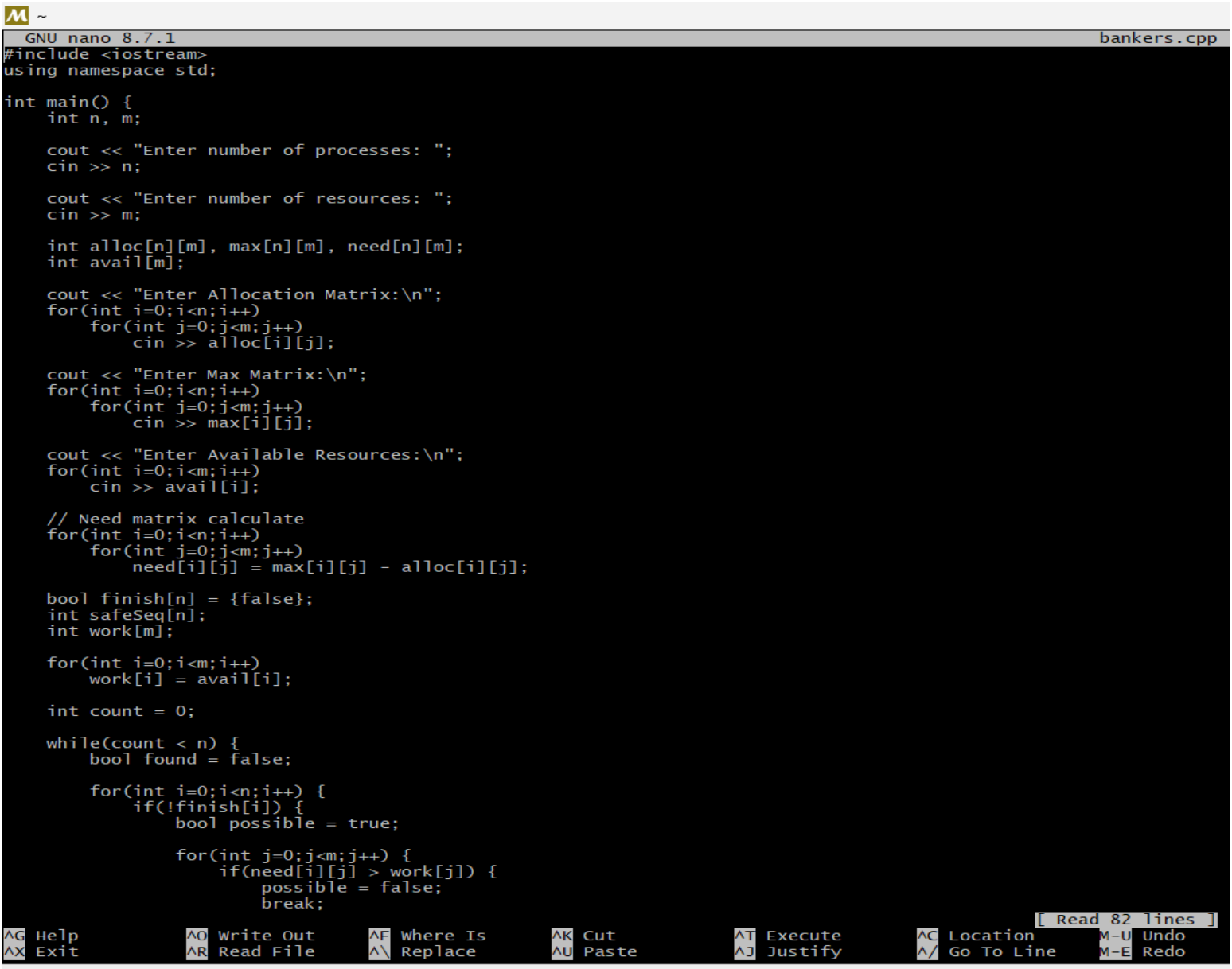
Practical 6th

Aim: A computer system has four different types of resources (printer, ROM, hard disk and RAM) and it needs to complete execution of five processes (Google Drive, Firefox, Word Processor, Excel and PowerPoint)' The number of allocated resources, maximum needed resources to complete the execution and total available resources should be taken from the user.

Write a program to :

- Calculate current need of resources by each process Find whether these processes can be executed with available resources. lf yes, show the correct order of process execution.

Code:



```
                for(int j=0;j<m;j++) {
                    if(need[i][j] > work[j]) {
                        possible = false;
                        break;
                    }
                }

                if(possible) {
                    for(int j=0;j<m;j++)
                        work[j] += alloc[i][j];

                    safeSeq[count++] = i;
                    finish[i] = true;
                    found = true;
                }
            }
        }

        if(!found) {
            cout << "System is NOT in safe state";
            return 0;
        }
    }

    cout << "System is in SAFE state\n";
    cout << "Safe sequence: ";

    for(int i=0;i<n;i++)
        cout << "P" << safeSeq[i] << " ";

    return 0;
}
```

```
^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo
```

Output:

```
priti@LAPTOP-US9V8OAL UCRT64 ~
$ ./bankers
Enter number of processes: 5
Enter number of resources: 4
Enter Allocation Matrix:
0 0 1 4
0 6 3 2
0 0 1 2
1 0 0 0
1 3 5 4
Enter Max Matrix:
0 6 5 6
0 6 5 2
0 0 1 2
1 7 5 0
2 3 5 6
Enter Available Resources:
1 6 2 0
System is in SAFE state
Safe sequence: P1 P2 P3 P4 P0
priti@LAPTOP-US9V8OAL UCRT64 ~
$
```